

Wi-Fi/BLE Module Specification

HM-WF7201



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1 Electrical Characteristics

Table 3-1 Electrical Parameters List

Parameter	Condition	Min.	Typ.	Max.	Unit
Module Model	HM-WF7201				
Package	Three-row Castellated Holes				
Dimensions	(20.3±0.10)x(15.3±0.10)x(2.60±0.10) With a shielding cover				mm
SPI Flash	Built-in 2MB				
Supported Interfaces	UART/HSPI/I2C/I2S/GPIO/PWM				
I/O Pins	18				
UART Rate	2	300	115200	600000	bps
Antenna Type	On-board PCB Antenna				
Frequency Range	802.11 b/g/n	2412		2484	MHz
Channel Bandwidth	802.11 b/g/n	20	20	20	MHz
Receive Sensitivity	1 Mbps DSSS		-99		dBm
	2 Mbps DSSS		-96		dBm
	11 Mbps DSSS		-90		dBm
	6 Mbps OFDM		-92		dBm
	54 Mbps OFDM		-76		dBm
	MCS0		-92		dBm
	MCS7		-74		dBm
Maximum Transmit Power	1 Mbps DSSS		19		dBm
	11 Mbps DSSS		19		dBm
	6 Mbps OFDM		20		dBm
	54 Mbps OFDM		17		dBm
	MCS0		19		dBm
	MCS7		16		dBm
Maximum Receive Level @ 2.4 GHz	5.5, 11 Mbps (8% PER, 1024 octets)			10	dBm
	6 ~ 54 Mbps (10% PER, 1024 octets)			5	dBm
	MCS0~7 (10% PER, 4096 octets)			2	dBm
BLE Specification	Operating Frequency Range	2402	-	2480	MHz
	Maximum Output Power	6	8	10	dBm

	Receive Sensitivity 30.8% PER		-98		dBm
Operating Temperature	-	-40		+8 5	°C
Operating Voltage	VCC	2.7	3.3	3. 6	V

2 Product Overview

The HM-WF7201 is a Wi-Fi/BLE wireless communication module based on the 2.4 GHz frequency band. It is a highly integrated single-chip solution for Wi-Fi 802.11b/g/n and Bluetooth Low Energy (BLE), designed for applications requiring a compact footprint. It integrates a powerful 32-bit MCU and a comprehensive set of peripheral interfaces, making it an ideal choice for advanced Internet of Things (IoT) applications. Utilizing advanced design and process technologies, it provides high integration and low power consumption in an extremely small package, suitable for smart lighting, smart home, indoor positioning, and other complex IoT applications.

The HM-WF7201 Wi-Fi module features complete and self-contained 802.11 b/g/n WLAN network functionality, supporting STA/AP/AP+STA modes. The built-in cache memory enhances system performance and optimizes the storage system. It can function as a Wi-Fi adapter connected to a microcontroller-based design simply via a UART interface. The module supports standard IEEE802.11 b/g/n protocols and a complete TCP/IP stack. Users can utilize this module to add networking capabilities to existing devices or build standalone network systems.

The HM-WF7201 Wi-Fi module is equipped with an integrated PCB antenna and supports transparent data transmission over Wi-Fi and BLE using AT commands.

3 Module Features

- 32-bit MCU, main frequency up to 160 MHz
- Protocol Support: Wi-Fi, BLE
- On-chip Integration: 2MB FLASH and 288KB RAM
- UART firmware upgrade and JTAG debugging
- Rich MCU Peripherals: 1xSPI/2xUART/1xI2C/1xGDMA/6x32-bit PWM/ADC/6x32-bit timer/1x watchdog timer/1x temperature sensor/1x TRNG/1x RTC
- **Wi-Fi Features:**
 - 1X1 IEEE 802.11 b/g/n
 - 20 MHz channel bandwidth
 - Supports STBC
 - Supports STA, AP, AP+STA modes
 - Transmit power up to +19 dBm
 - Receive sensitivity down to -99 dBm
- **BLE Features:**
 - Bluetooth 5.2
 - Supports 1 Mbps, 2 Mbps, and Long Range (125 kbps and 500 kbps) data rates
 - Supports Advertising Extensions
 - Integrated Bluetooth LE/WLAN coexistence (PTA)
- **Clock Management:**
 - External Oscillator: 26 MHz crystal oscillator (X26M)
 - Internal Oscillators: 26~160 MHz Digitally Controlled Oscillator (DCO), 32 kHz Ring Oscillator (ROSC)
 - 480 MHz DPLL
- **Power Management:**

- Supports 2.7 to 3.6 V VBAT
- Supports on-chip Power-On Reset (POR) and Brown-out Detector (BOD)
- Embedded LDO regulator

4 Module Function Description

The HM-WF7201 module integrates two wireless communication protocols, Wi-Fi and BLE. It can configure both wireless networks using the AT command set according to different application scenarios, enabling the transmission and reception of data over Wi-Fi and BLE.

For the Wi-Fi function, it can be configured in STA mode to connect to other APs (e.g., a home router), or in AP mode to provide a wireless network for other STAs. It also supports a concurrent AP+STA mode. Based on the Wi-Fi network connection, various types of TCP/IP services can be established, such as TCP server, TCP client, and UDP client. These services enable the sending and receiving of user data and Wi-Fi transparent transmission. Specific implementation details can be found in the subsequent Wi-Fi and TCP/IP AT command sections.

For the BLE function, it supports Master/Slave modes, including BLE scanning, advertising, connection establishment, and data transmission. These functions allow for sending and receiving user data. Specific implementation details can be found in the subsequent BLE AT command section.

5 Application Schematic

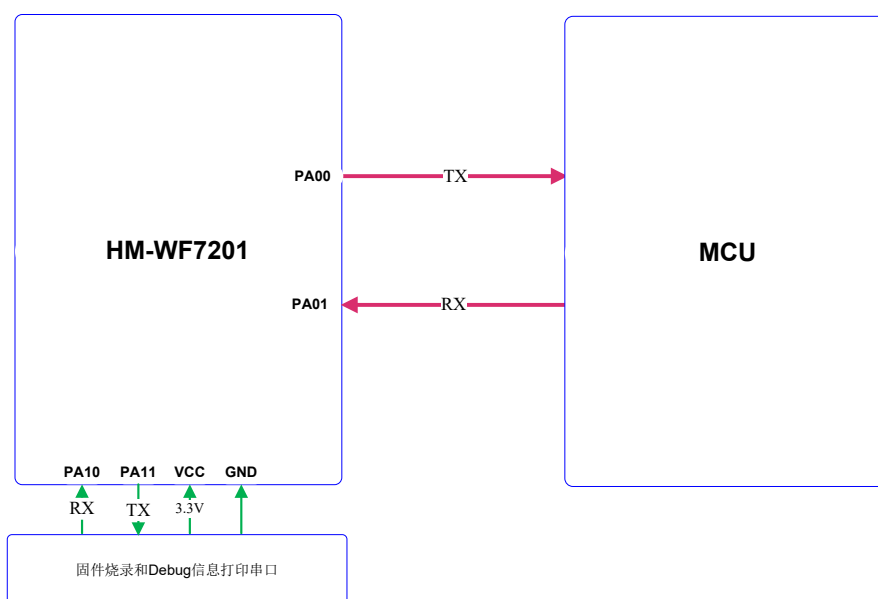


Figure 5.1 Module Application Schematic (Communication with external device via UART)

The HM-WF7201 module communicates with the external MCU via the serial port. The default baud rate of the serial port is 115200 and it is powered by 3.3V. As shown in Figure 5.1, the module can be connected via the serial port to send the AT command set, completing basic configuration of the module, parameter configuration of Wi-Fi/BLE, establishing TCP/IP services, sending or receiving user data, and upgrading the module firmware, etc.

6 Module Pins

6.1 Module Pin Layout

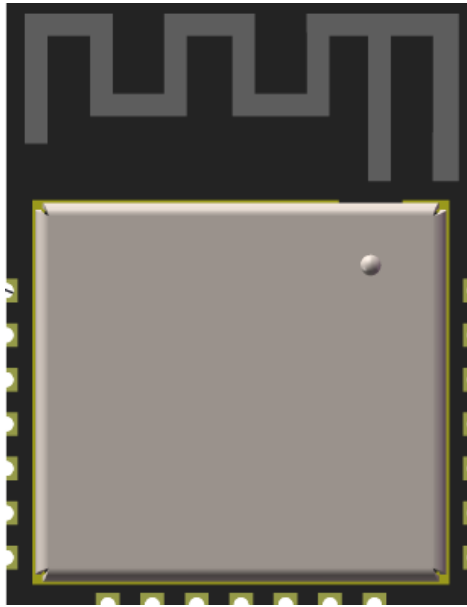


Figure6.1 Module Pin Layout Diagram (Top View)

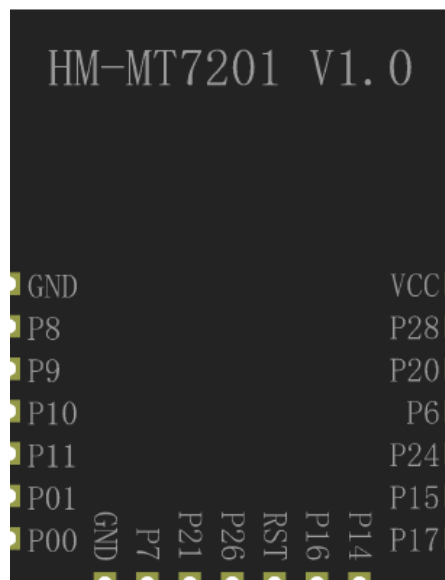


Figure6.2 Module Pin Layout Diagram (Bottom View)

6.2 Module Pin Definition

Table 6.1 Module Pin Definition

Pin No.	Pin Name	Type	Description
1	VCC	DV	VCC 3.3V POWER SUPPLY
2	P28	I/O	P28/ADC4
3	P20	I/O	P20/ADC3
4	P06	I/O	PA06/CLK13M/PWM0/TCK/ANT0
5	P24	I/O	P24/LPO_CLK/PWM4/ANTLSB/SCL/ADC2
6	P15	I/O	P15/CSN/ANT1/SCL
7	P17	I/O	P17/MISO/ANT3/SDA
8	P14	I/O	P14/SCK/ANT0
9	P16	I/O	P16/MOSI/ANT2
10	RST	I/O	Reset (Active Low)
11	P26	I/O	P26/PWM5/ANTMSB/SDA/ADC1
12	P21	I/O	PA21 (during system restart, grounding P21 enters ATE mode)
13	P07	I/O	P07/PWM1/TMS/ANT1
14	GND	DG	Ground
15	P00	I/O	P00/TXEN/UART2-TX
16	P01	I/O	P01/RXEN/ADC5/UART2-RX
17	P11	I/O	PA11/UART1-TX
18	P10	I/O	PA10/ADC6/UART1-RX
19	P09	I/O	P09/PWM3/TDO/ANT3
20	P08	I/O	P08/PWM2/TDI/ANT2/CLK26M
21	GND	DG	Ground

Note:

1. During development, avoid grounding the P21 pin during system restart, as this will prevent the normal system from starting.

2. UART1 is the firmware flashing UART (unchangeable) and is configured as the debug information printing UART by default. UART2 is configured as the communication UART by default, used for connecting to an external MCU.

7 -Module Dimensions

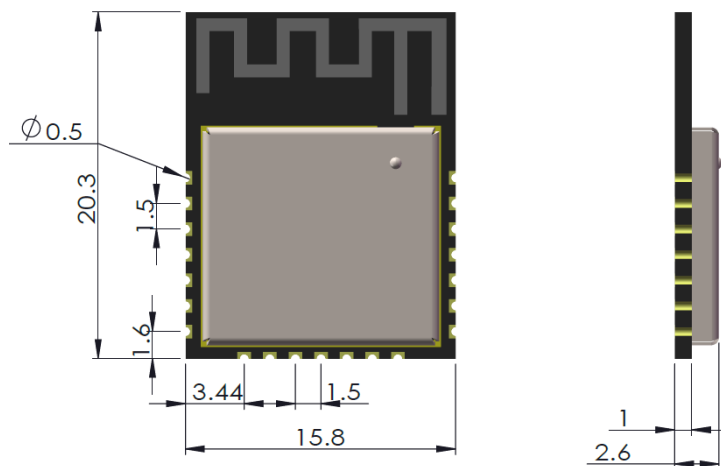


Figure7.1 HM-WF7201 Module Dimension Diagram with Shield Can (Unit: mm)

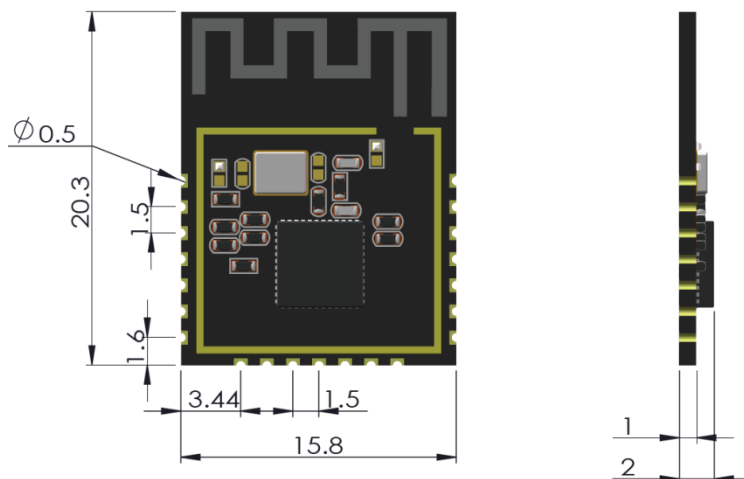


Figure7.2 HM-WF7201 Module Dimension Diagram without Shield Can (Unit: mm)

8 AT Command Set

8.1 AT Command Description

AT commands generally consist of three parts: a command field, a parameter field, and a carriage return and line feed ending field. For example, setting UART parameters: AT+UART_CUR=115200,8,1,0,0rn.

Command Format	Description
CMD=<param1>,<param2>...	Parameters in <> like param1, param2 are mandatory
CMD=[param1],[param2]...	Parameters in [] like param1, param2 are optional.
CMD={param1 param2}...	Parameters in {} like param1, param2 are optional, and one must be selected.

Note:

- A comma must exist between the command and parameters, and between parameters themselves.
- Please set the required number of parameters correctly; too many or too few parameters will cause software parsing to fail.
- AT commands are strictly case-sensitive. Failure to follow this document's requirements may prevent correct software parsing.
- The default UART configuration parameters are 115200 bps baud rate, 8 data bits, no parity, 1 stop bit.
- For the CRLF termination field, please enable the "Newline" or "CR+LF" function in your serial terminal tool.

8.2 AT Command Response Indications

After a user inputs an AT command in a serial debugging tool, the software will return a corresponding parsed status information, regardless of whether the command was correct or not. Specific descriptions are in the table below:

Return Status	Indication	Description
---------------	------------	-------------

Success	OK	AT command processing completed, returns OK.
Failure	ERROR	AT command error or error occurred during execution.

8.3 Basic AT Command Set

Command	Function Description
AT	AT command ready check.
AT+HELP?	Help information for AT commands.
AT+GMR?	Display firmware version information.
AT+RST	System restart.
AT+GSLP	Enter Deep-sleep mode.
AT+WORKMODE?	Query device working mode.
AT+WORKMODE	Set device working mode.
ATE1/ATE0	Enable/Disable AT command echo.
AT+RESTORE	Restore device to factory settings.
AT+USRRAM?	Query available RAM space.
AT+SYSFLASH?	View device partition table.
AT+SYSTIMESTAMP?	Query device system time.
AT+SYSTIMESTAMP	Set device system time.
AT+SLEEPWCFG	Set device wake-up source and wake-up GPIO.
AT+SYSSTORE?	Query storage mode for AT command configuration.
AT+SYSSTORE	Set storage mode for AT command configuration.
AT+UART_DEF?	Query UART configuration information (stored in Flash).
AT+UART_DEF	Configure UART and save to Flash.
AT+UART_CUR	Configure UART without saving to Flash.
AT+PRODUCTID?	Query device product ID.
AT+PRODUCTID	Set device product ID.
AT+DEVICENAME?	Query device name.
AT+DEVICENAME	Set device name.
AT+REGION?	Query device country code.
AT+REGION	Set device country code.

AT Command

Command	AT
Response	OK
Function	Test if the AT command execution environment is ready.
Parameter Description	None
Example	AT OK
Notes	None

AT+HELP? Command

Command	AT+HELP?
Response	Returns help information for supported AT commands.
Function	View supported AT commands.
Parameter Description	None
Example	AT+HELP? CMDRSP:cmd:AT+HELP?,help:AT+HELP? CMDRSP:cmd:AT+GMR,help:AT+GMR CMDRSP:cmd:AT+RST,help:AT+RST CMDRSP:cmd:AT+GSLP,help:AT+GSLP OK
Notes	None

AT+GMR? Command

Command	AT+GMR?
Response	Returns firmware version information.
Function	Get firmware version information.
Parameter Description	None
Example	AT+GMR? <ATVER:1.1> <SDKVER:3.0.78> OK
Notes	None

AT+RST Command

Command	AT+RST
Response	OK
Function	System restart.
Parameter Description	None
Example	AT+RST OK
Notes	None

AT+GSLP Command

Command	AT+GSLP=<time>
Response	Returns OK on success; returns error code on failure.
Function	Enter Deep-sleep mode.
Parameter Description	time: Time to enter deep sleep mode, in milliseconds.
Example	AT+GSLP=5000 OK
Notes	None

AT+WORKMODE? Command

Command	AT+WORKMODE?
Response	Return to the current working mode. 0: Factory mode; 1: Normal mode, only supports AT commands.
Function	Query the current working mode of the device.
Parameter Description	None.
Example	AT+WORKMODE? +WORKMODE:0 OK
Notes	None

AT+WORKMODE Command

Command	AT+WORKMODE=<mode>
----------------	--------------------

Response	Response: If successful, return "OK"; if failed, return the error code.
Function	Function: Set the working mode of the device.
Parameter Description	Parameter Description: mode: 0: Factory mode; 1: Normal mode, only supports AT commands.
Example	AT+WORKMODE=0 OK
Notes	None

ATE1/ATE0 Command

Command	ATE1 or ATE0
Response	OK
Function	Enable/Disable AT command echo.
Parameter Description	None
Example	ATE0 OK
Notes	None

AT+RESTORE Command

Command	AT+RESTORE
Response	OK
Function	Restore device to factory reset.
Parameter Description	None
Example	AT+RESTORE OK
Notes	None

AT+USRRAM? Command

Command	AT+USRRAM?
Response	Returns the currently available RAM space.
Function	Query available RAM space.

Parameter Description	None
Example	AT+USRRAM? +USERRAM:127416 OK
Notes	None

AT+SYSFLASH? Command

Command	AT+SYSFLASH?
Response	Returns the currently available RAM space.
Function	Query available RAM space.
Parameter Description	None
Example	AT+SYSFLASH? <pre> 0 Bootloader Dev:0 0x00000000 0x0000f000 1 Application Dev:0 0x00011000 0x00121000 2 ota Dev:0 0x00132000 0x000ae000 3 RF Firmware Dev:0 0x001e0000 0x00001000 4 NET info Dev:0 0x001e1000 0x00001000 5 USER info Dev:0 0x001ee000 0x00001000 6 BLE SERVER Dev:0 0x001ef000 0x00001000 ... ble bonding info Dev:0 0x001f0000 0x00001000 </pre> OK
Notes	None

AT+SYSTIMESTAMP? Command

Command	AT+SYSTIMESTAMP?
Response	Returns the device's system time.
Function	Query the device's system time.
Parameter Description	None
Example	AT+SYSTIMESTAMP? +SYSTIMESTAMP:4171 OK
Notes	None

AT+SYSTIMESTAMP Command

Command	AT+SYSTIMESTAMP=<TIMESTAMP>
Response	Returns OK on success; returns error code on failure.
Function	Set the device's system time.
Parameter Description	TIMESTAMP: Unix time, in seconds.

Example	AT+SYSTIMESTAMP=1686305483 OK
Notes	None

AT+SLEPPWCFG Command

Command	AT+SLEPPWCFG=<wakeup source>,<param1>[,<param2>]
Response	Returns OK on success; returns error code on failure.
Function	Set device wake-up source and wake-up GPIO.
Parameter Description	<wakeup source>: Wake-up source 0: Timer wake-up 1: Reserved configuration 2: GPIO wake-up <param1>: - When wakeup source is timer, this parameter specifies sleep time in seconds. - When wakeup source is GPIO, this parameter defines the GPIO pin number. <param2>: (Used when wakeup source is GPIO) +0: Rising edge trigger +1: Falling edge trigger
示例	AT+SLEPPWCFG=0,2 // Auto-wake after 2 seconds <pre>[11:25:52.210]收<◆AT+SLEPPWCFG=0,2 OK [11:25:54.223]收<◆BK7238_1.0.15 REG:opsr spsr r13 r14 SVC:0x000000D3 0xB4D4DFE1 0xBAD26182 IRQ:0x000000D2 0x00000010 0x07E40D8B 0xAD01F12B FIR:0x000000D1 0x00000010 0xCE6AF6DB 0xE70D8703 SYS:0x000000DF 0xAE02CCD4 0xFCC95E4B ST:0x00000002 [32:22m[I/FAL] Fal(V0.4.0)success[0m [36:22m[I/OTA] RT-Thread OTA package(V0.2.8-beken-11332824-20220604) initialize success.[0m go os_addr(0x10000)..... [11:25:54.828]收<◆ ready</pre> AT+SLEPPWCFG=2,20,0 // Wake-up on GPIO20 rising edge OK
Notes	None

AT+SYSSTORE? Command

Command	AT+SYSSTORE?
Response	Returns the storage mode for AT command configuration. 0: Command configuration not saved to flash; 1: Command configuration saved to flash (default).

Function	Query the storage mode for AT command configuration.
Parameter Description	None
Example	AT+SYSSTORE? +SYSSTORE:1 OK
Notes	None

AT+SYSSTORE Command

Command	AT+SYSSTORE=<store mode>
Response	Returns OK on success; returns error code on failure.
Function	Set the storage mode for AT command configuration.
Parameter Description	store mode: 0: Command configuration not saved to flash; 1: Command configuration saved to flash (default).
Example	AT+SYSSTORE=1 OK
Notes	None

AT+UART_DEF? Command

Command	AT+UART_DEF?
Response	Returns the current UART configuration information.
Function	Query UART configuration information.
Parameter Description	None
Example	AT+UART_DEF? +UART_DEF:115200,8,1,0,0 OK
Notes	None

AT+UART_DEF Command

Command	AT+UART_DEF:<baudrate>,<databits>,<stopbits>,<parity>,<flow control>
Response	Returns OK on success; returns error code on failure.
Function	Configure UART and save to Flash.

Parameter Description	<baudrate>: UART baud rate, range: 9600 ~ 2000000. <databits>: Data bits: 5:5 bit ; 6:6 bit ; 7:7 bit ; 8:8 bit. <stopbits>: Stop bits: 1, 2. <parity>: Parity: 0: None, 1: Odd, 2: Even. <flow control>: Flow control: 0: Disable, 1: Enable RTS, 2: Enable CTS, 3: Enable RTS and CTS.
Example	AT+UART_DEF=115200,8,1,0,0 OK
Notes	None

AT+UART_CUR Command

Command	AT+UART_CUR=<baudrate>,<databits>,<stopbits>,<parity>,<flow control>
Response	Returns OK on success; returns error code on failure.
Function	Configure UART without saving to Flash.
Parameter Description	<baudrate>: UART baud rate, range: 9600 ~ 2000000. <databits>: Data bits: 5, 6, 7, 8. <stopbits>: Stop bits: 1, 2. <parity>: Parity: 0: None, 1: Odd, 2: Even. <flow control>: Flow control: 0: Disable, 1: Enable RTS, 2: Enable CTS, 3: Enable RTS and CTS.
Example	AT+UART_CUR=115200,8,1,0,0 OK
Notes	None

AT+PRODUCTID? Command

Command	AT+PRODUCTID?
Response	Returns the product ID.
Function	Query device product ID.
Parameter Description	None
Example	AT+PRODUCTID? +PRODUCTID:PP1234 OK
Notes	None

AT+PRODUCTID Command

Command	AT+PRODUCTID=<productid>
Response	Returns OK on success; returns error code on failure.
Function	Set the product ID of the equipment.
Parameter Description	<productid>: Product ID, maximum length of 64 bytes
Example	AT+PRODUCTID=PP1234 OK
Notes	None

AT+DEVICENAME? Command

Command	AT+DEVICENAME?
Response	Returns the device name.
Function	Query the device name.
Parameter Description	None
Example	AT+DEVICENAME? +DEVICENAME:BK7238_01 OK
Notes	None

AT+DEVICENAME Command

Command	AT+DEVICENAME=<device name>
Response	Returns OK on success; returns error code on failure.
Function	Set the device name.
Parameter Description	device name: Device name, maximum length 64 bytes.
Example	AT+DEVICENAME=BK7238_01 OK
Notes	None

AT+REGION? Command

Command	AT+REGION?
Response	Returns the device country code.
Function	Query the device country code.
Parameter Description	None
Example	AT+REGION? +REGION:China OK
Notes	None

AT+REGION Command

Command	AT+REGION=<COUNTRY>
Response	Returns OK on success; returns error code on failure.
Function	Set the device country code.
Parameter Description	<COUNTRY>: Country where the device is located, maximum length 64 bytes.
Example	AT+REGION=China OK
Notes	None

8.4 Wi-Fi Command Set

Command	Function Description
AT+CWMODE?	Query Wi-Fi working mode.
AT+CWMODE	Set Wi-Fi working mode.
AT+CIPSTAMAC?	Query device MAC address.
AT+CIPSTAMAC	Set device MAC address.
AT+CWSAP	Create an AP (Access Point).
AT+CWQIF	Stop an AP.
AT+CIPAP?	Query AP IP address and gateway.
AT+CIPAP	Set AP IP address and gateway.
AT+CWLIF?	Query stations connected to the AP.
AT+CWLAP	Scan for available APs nearby.
AT+CWDHCP?	Query DHCP status.
AT+CWDHCP	Set DHCP mode.
AT+CWJAP	Create a STA and connect to a specified AP.
AT+CWQAP	STA disconnects from the current AP.
AT+CWJAP?	STA reconnects to the current AP.
AT+CIPSTA?	Query STA IP information.
AT+CIPSTA	Configure STA IP information.
AT+CIFSR?	Query module address information.
AT+STAST?	Query STA status.

AT+CWMODE? Command

Command	AT+CWMODE?
Response	Returns the Wi-Fi working mode.
Function	Query Wi-Fi working mode.
Parameter Description	None
Example	AT+CWMODE? msg: AT+CWMODE? +WLMODE:1,1 OK
Notes	None

AT+CWMODE Command

Command	AT+CWMODE=<mode>[,<auto connect>]
Response	Returns OK on success; returns error code on failure.
Function	Set Wi-Fi working mode.
Parameter Description	<mode>: Mode: 0: No Wi-Fi mode, disable Wi-Fi; 1: STA mode; 2: SoftAP mode; 3: AP+STA mode. <auto connect>: 0: Disable auto-connect to AP; 1: Enable auto-connect to AP (if AP configuration is saved, it will auto-connect on reboot).
Example	AT+CWMODE=1,1 OK
Notes	None

AT+CIPSTAMAC? Command

Command	AT+CIPSTAMAC?
Response	Returns the device MAC address.
Function	Query device MAC address.
Parameter Description	None
Example	AT+CIPSTAMAC? +WLMAC:c8:47:8c:42:88:50 OK
Notes	None

AT+CIPSTAMAC Command

Command	AT+CIPSTAMAC=<mac>
Response	Returns OK on success; returns error code on failure.
Function	Set device MAC address.
Parameter Description	<mac>: Supports "MM:MM:MM:MM:MM:MM" and "MM-MM-MM-MM-MM-MM" formats.
Example	AT+CIPSTAMAC=c8:47:8c:42:88:50 OK
Notes	Must be executed before configuring Wi-Fi parameters.

AT+CWSAP Command

Com mand	AT+CWSAP=<SSID>,<PWD>,<channel>,<ECN>,<max conn>,<hidden>
Resp onse	Returns OK on success; returns error code on failure.
Funct ion	Create a Soft AP.
Para meter Descr iption	<SSID>: String parameter, access point name. <PWD>: String parameter, password, range: 8 ~ 64 bytes, ASCII. <channel>: Channel number. <proto>: 0: 802.11bgn standard; 1: 802.11bg standard; 2: 802.11b standard. <ECN>: Encryption method, WEP not supported: 0 OPEN; 1 WEP; 2 WPA_TKIP; 3 WPA_AES; 4 WPA2_TKIP; 5 WPA2_AES; 6 AES_TKIP_WPA2_MIXED; 7 WPA3_SAE; 8 WPA3_WPA2 (default); 9 EAP; 10 OWE; 11 AUTO. [<max conn>]: Maximum number of STAs allowed to connect to Soft AP, range: [1,3]. [<hidden>]: 0: Broadcast SSID (default); 1: Do not broadcast SSID.
Exam ple	AT+CWSAP=aaabkpxm01,12345678,10,8,3,0 OK
Notes	Related to CWMODE. Can only configure Soft AP when CWMODE mode is set to 2 or 3.

AT+CWQIF Command

Command	AT+CWQIF
Response	Returns OK on success; returns error code on failure.
Function	Stop the enabled AP.
Parameter Description	None
Example	AT+CWQIF OK
Notes	Related to CWMODE. Can only use this command when CWMODE mode is set to 2/3.

AT+CIPAP? Command

Command	AT+CIPAP?
Response	Returns the AP IP address and gateway.

Function	Query AP IP address and gateway.
Parameter Description	None
Example	AT+CIPAP? +CIPAP:192.168.42.1,255.255.255.0,192.168.42.1 OK
注意事项	Related to CWMODE, this instruction can only be used when the mode configuration in CWMODE is set to 2/3.

AT+CIPAP Command

Command	AT+CIPAP=<IP>,<MASK>,<GW>
Response	Returns OK on success; returns error code on failure.
Function	Set AP IP address and gateway.
Parameter Description	<IP>: String parameter, IPv4 address. <MASK>: Subnet mask. <GW>: Gateway IPv4 address.
Example	AT+CIPAP=192.168.42.1,255.255.255.0,192.168.42.1 OK
Notes	Related to CWMODE. Can only use this command when CWMODE mode is set to 2/ 3.

AT+CWLIF? Command

Command	AT+CWLIF?
Response	Returns information about STAs associated with the AP.
Function	Query STAs connected to the AP.
Parameter Description	None
Example	AT+CWLIF? +CWLIF:index:0,ip:192.168.42.100,mac:c8:47:8c:42:88:50, rssi:-36 OK
Notes	Related to CWMODE. Can only use this command when CWMODE mode is set to 2/3.

AT+CWLAP Command

Command	AT+CWLAP
Response	Returns information about scanned APs.
Function	Scan for available APs nearby.

Parameter Description	None
Example	AT+CWLAP <pre>[17:23:37.711]H+>+CWLAP:(3,"aashkpm01",-33,"c8:47:8c:42:88:48",11) +CWLAP:(3,"www.hoperf.cn",-37,"7c:b5:9b:94:58:0a",11) +CWLAP:(3,"aaa_asr_2g",-37,"f4:84:8d:36:65:e3",13) +CWLAP:(3,"",-38,"f8:84:8d:36:65:e3",13) +CWLAP:(3,"HP-matter-001-56",-49,"02:2b:7b:67:bc:c3",9) +CWLAP:(3,"HP-matter-001",-50,"02:2b:7b:67:bc:c2",9) +CWLAP:(0,"TPQuest_D9E2",-51,"9e:97:cc:fd:d9:e2",1) +CWLAP:(3,"TP-LINK_D9E2",-51,"9e:97:cc:fd:d9:e2",1) +CWLAP:(3,"",-51,"a2:97:cc:fd:d9:e2",1) +CWLAP:(3,"NT2",-52,"20:76:93:4a:5f:52",12) +CWLAP:(3,"www.hoperf.cn",-52,"7c:b5:9b:94:58:0a",6) +CWLAP:(3,"Gateway_F6M4ID",-52,"24:e1:24:f6:d4:1d",6) +CWLAP:(3,"HUAWEI-H66YZCO",-54,"5e:7f:c7:36:a2:11",3) +CWLAP:(3,"DIRECT-38-HUAWEI PixLab X1",-61,"54:55:d5:12:d9:75",11) +CWLAP:(3,"Mate 40 Pro",-63,"4e:31:0e:c3:c1:83",2) +CWLAP:(3,"IPC_TMS",-68,"b4:6d:c2:db:4d:7c",4) +CWLAP:(3,"HUAWEI_C336",-69,"62:38:a2:36:c3:18",10) +CWLAP:(3,"pudayaof@HUAWEI",-70,"36:96:45:c4:4a:f3",4) +CWLAP:(3,"DIRECT-6A-HP Laser 1188pwn",-74,"7e:4d:8f:7d:94:6a",1) +CWLAP:(3,"HP-OP_Wi-Fi5",-75,"62:3a:b2:6d:26:6b",1) +CWLAP:(3,"HoperF30",-79,"02:57:44:3b:fe:bf",11)</pre> OK
Notes	Related to CWMODE. Can only use this scan function when CWMODE mode is set to STA

AT+CWDHCP? Command

Command	AT+CWDHCP?
Response	Returns the current DHCP status of the device.
Function	Query DHCP status.
Parameter Description	None
Example	AT+CWDHCP? +CWDHCP:2 OK
Notes	None

AT+CWDHCP Command

Command	AT+CWDHCP=<enable>,<mode>
Response	Returns OK on success; returns error code on failure.
Function	Set DHCP mode.
Parameter Description	<operate>: 0: Disable; 1: Enable. <mode>: Bit0: DHCP for STA; Bit1: DHCP for AP. <state> DHCP state: Bit0: 0: Disable STA DHCP; 1: Enable STA DHCP. Bit1: 0: Disable AP DHCP; 1: Enable AP DHCP.
Example	AT+CWDHCP=0,1 // Disable STA DHCP AT+CWDHCP=0,2 // Disable AP DHCP AT+CWDHCP=0,3 // Disable STA+AP DHCP AT+CWDHCP=1,1 // Enable STA DHCP AT+CWDHCP=1,2 // Enable AP DHCP AT+CWDHCP=1,3 // Enable STA+AP DHCP

Notes	Related to CWMODE. Configure the matching CWDHCP mode according to the mode in CWMODE.
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AT+CWJAP Command

Command	AT+CWJAP=[<SSID>],[<PWD>],[<bssid>],[<pci_en>],[<reconn_interval>],[<listen_interval>],[<scan_mode>],[<jap_timeout>],[<pmf>]
Response	Returns OK on success; returns error code on failure.
Function	Create a STA and connect to a specified AP.
Parameter Description	<SSID>: SSID of the AP. <PWD>: Password of the AP. <bssid>: BSSID of the AP.
Example	AT+CWJAP=aaabkpxm01,12345678 OK
Notes	Related to CWMODE. Can only use this command when CWMODE mode is set to STA.

AT+CWQAP Command

Command	AT+CWQAP
Response	Returns OK on success; returns error code on failure.
Function	STA disconnects from the current AP.
Parameter Description	None
Example	AT+CWQAP OK
Notes	None

AT+CWJAP? Command

Command	AT+CWJAP?
Response	Returns OK on success; returns error code on failure.
Function	STA reconnects to the current AP.
Parameter Description	None
Example	AT+CWJAP? OK

Notes	After executing AT+CWQAP to disconnect from the current AP, this command can be executed to restore the connection with the current AP.
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AT+CIPSTA? Command

Command	AT+CIPSTA?
Response	Returns STA IP information.
Function	Query STA IP information.
Parameter Description	None
Example	AT+CIPSTA? +CIPSTA:ip:192.168.42.100 +CIPSTA:gateway:192.168.42.1 +CIPSTA:netmask:255.255.255.0 OK
Notes	Related to CWMODE. Can only use this command when CWMODE mode is set to STA.

AT+CIPSTA Command

Command	AT+CIPSTA =<IP>,<MASK>,<GW>
Response	Returns OK on success; returns error code on failure.
Function	Configure STA IP information.
Parameter Description	<IP>: String parameter, IPv4 address. <MASK>: Subnet mask. <GW>: Gateway IPv4 address.
Example	AT+CIPSTA=192.168.42.1,255.255.255.0,192.168.42.1 OK
Notes	Related to CWMODE. Can only use this command when CWMODE mode is set to STA.

AT+CIFSR? Command

Command	AT+CIFSR?
Response	Returns the module's address information.
Function	Query the module's address information.
Parameter Description	None

Example	AT+CIFSR? +CIFSR:STAIP,"192.168.42.100" +CIFSR:STAMAC,"c8:47:8c:42:88:50" OK
Notes	None

AT+STAST? Command

Command	AT+STAST?
Response	Returns the status of the STA.
Function	Query STA status.
Parameter Description	None
Example	AT+STAST? +STAST:connected,dhcp=0,ip=192.168.42.100,gate=192.168.42.1. mask=255.255.255.0, mac=c8:47:8c:42:88:50. RSSI:-25, ssid:aaabkpxm01,channel:11, security:6 OK
Notes	Related to CWMODE. Can only use this command when CWMODE mode is set to STA.

8.5 TCP/IP Command Set

Command	Function Description
AT+CIPSTATUS?	Query TCP/UDP/SSL connection status information.
AT+CIPSERVER	Create a TCP Server.
AT+CIPSTART	Establish a TCP/UDP/TLS connection.
AT+CIPCLOSE	Close a connection.
AT+CIPMODE	Set data transmission mode.
AT+CIPMUX	Set connection mode.
AT+CIPMUX?	Query current connection mode.
AT+CIPSEND	Send data in normal/transparent transmission mode.
AT+CIPSSLCPK?	Query SSL client PSK.
AT+CIPSSLCPK	Set SSL client PSK.
AT+CIPSNTPCFG?	Query NTP server.
AT+CIPSNTPCFG	Set NTP server.
AT+CIPSNTPTIME?	Query SNTP time.
AT+CIPDOMAIN	Resolve domain name.
AT+CIPDNS	Set DNS server.

AT+CIPSTATUS? Command

Command	AT+CIPSTATUS?
Response	Returns status information for various connections.
Function	Query TCP/UDP/SSL connection status information.
Parameter Description	None
Example	AT+CIPSTATUS? +BIPSTATUS:0,connected, 1, 192.168.42.1, 333,333,1 OK
Notes	None

AT+CIPSERVER Command

Command	AT+CIPSERVER=<mode>[,port]
Response	Returns OK on success; returns error code on failure.
Function	Create a TCP Server.

Parameter Description	<mode>: 0: Use default port: 333; 1: Specify port number. [,port]: When mode==1, specify the port number.
Example	AT+CIPSERVER=1,8899 // Create TCP Server on port 8899 OK
Notes	None

AT+CIPSTART Command

Command	AT+CIPSTART=[Link ID],<type>,<remote host>,<remote port> [,<keepalive>,<local port>],<local ip>
Response	Returns OK on success; returns error code on failure.
Function	Establish a TCP/UDP/TLS connection.
Parameter Description	<Link ID>: Network connection ID (0~5), supports up to 6 connections. <type>: Connection type: 0: TCP, 1: UDP, 2: TLS, 3: DTLS. <remote host>: Remote connection IP. <remote port>: Remote connection port. <keepalive>: TCP keep-alive interval, default: 0; 0: Disable TCP keep-alive, 1 ~ 7200: Detection interval in seconds.
Example	AT+CIPMUX=0 // Enable single connection mode AT+CIPSTART=tcp,192.168.42.1,8899 // Create TCP connection with ID 0 AT+CIPMUX=1 // Enable multiple connection mode AT+CIPSTART=2,udp,192.168.42.1,333,333,0 // Create UDP connection with ID 2
Notes	[link ID]: In single connection mode (AT+CIPMUX=0), Link ID uses default value 0 and this parameter can be omitted.

AT+CIPCLOSE Command

Command	AT+CIPCLOSE=<link ID>
Response	Returns OK on success; returns error code on failure.
Function	Close a connection.
Parameter Description	<Link ID>: Established network connection ID (0~5), supports up to 6 connections.
Example	AT+CIPCLOSE=0 OK
Notes	None

AT+CIPMODE Command

Command	AT+CIPMODE=<mode>
Response	Returns OK on success; returns error code on failure.
Function	Set data transmission mode.
Parameter Description	<mode>: 0: Normal data transmission mode; 1: Data transparent transmission mode.
Example	AT+CIPMODE=1 // Set to data transparent transmission mode. OK
Notes	When using data transparent transmission mode (AT+CIPMODE=1), the connection must be in single connection mode (AT+CIPMUX=0).

AT+CIPMUX Command

Command	AT+CIPMUX=<mode>
Response	Returns OK on success; returns error code on failure.
Function	Set connection mode.
Parameter Description	<mode>: 0: Single connection mode; 1: Multiple connection mode.
Example	AT+CIPMUX=0 // Set to single connection mode OK
Notes	None

AT+CIPMUX? Command

Command	AT+CIPMUX?
Response	Returns the current connection mode.
Function	Query current connection mode.
Parameter Description	None
Example	AT+CIPMUX? +CIPMUX:0 OK
Notes	None

AT+CIPSEND Command

Command	AT+CIPSEND=[<link ID>,<length>,<remotehost>,<remoteport>]
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Response	Returns OK on success; returns error code on failure.
Function	Send data in normal/transparent transmission mode.
Parameter Description	<link ID>: Network connection ID (0 ~ 4), used for multiple connections. <length>: Data length, maximum: 2048 bytes. <remote host>: For UDP transmission, can specify the destination host: IPv4 address or domain name. <remote port>: For UDP transmission, can specify the destination port.
Example	<pre>AT+CIPSEND=0,10 // Normal mode, send 10 chars on link ID 0 1234567890 AT+CIPSEND=0,10 // Normal mode, send another 10 chars on link ID 0 0987654321 AT+CIPSEND //In transparent transmission mode, send data continuously 1234567890 0987654321 +++ // Exit transparent mode, Note: Exit transparent mode No "\r\n" after +++</pre>
Notes	AT+CIPMODE command sets the data transmission mode.

AT+PING Command

Command	AT+PING=<host>
Response	Returns OK on success; returns error code on failure.
Function	Ping a specified network device.
Parameter Description	<host>: IPv4 address or domain name.
Example	<pre>AT+PING=www.baidu.com 60 bytes from 183.240.99.169 icmp_seq=0 ttl=53 time=16 ticks 60 bytes from 183.240.99.169 icmp_seq=1 ttl=53 time=18 ticks 60 bytes from 183.240.99.169 icmp_seq=2 ttl=53 time=16 ticks 60 bytes from 183.240.99.169 icmp_seq=3 ttl=53 time=22 ticks OK</pre>
Notes	None

AT+CIPSSLCPSK? Command

Command	AT+CIPSSLCPSK?
Response	Returns SSL client PSK information.
Function	Query SSL client PSK.

Parameter Description	None
Examp	AT+CIPSSLCPK? +CIPSSLCPK:5,2fe2b750a610dab52fa24374c0cb90a2e52b4c02f947107f83f4c11d81f74681,1V2OAXDZ65AT&BK7238 OK
Notes	None

AT+CIPSSLCPK Command

Command	AT+CIPSSLCPK=<Link ID>,<PSK key>,<hint>
Response	Returns OK on success; returns error code on failure.
Function	Set SSL client PSK.
Parameter Description	<Link ID>: Connection ID (0~5). <PSK key>: PSK identity, maximum length: 48. <hint>: PSK hint, maximum length: 48.
Example	AT+CIPSSLCPK=5,2fe2b750a610dab52fa24374c0cb90a2e52b4c02f947107f83f4c11d81f74681,1V2OAXDZ65AT&BK7238 OK
Notes	For this configuration to take effect immediately, run this command before establishing an SSL connection.

AT+CIPSNTPCFG? Command

Command	AT+CIPSNTPCFG?
Response	Returns the current NTP server.
Function	Query the current NTP server.
Parameter Description	None
Example	AT+CIPSNTPCFG? +CIPSNTPCFG:1,8,cn.pool.ntp.org OK
Notes	None

AT+CIPSNTPCFG Command

Command	AT+CIPSNTPCFG=<enable>,<time zone>,<SNTP server>
Response	Returns OK on success; returns error code on failure.
Function	Set NTP server.

Parameter Description	<enable>: Set SNTP server: 1: Set SNTP server; 0: Do not set SNTP server. <time zone>: Numeric range: [-12,12], marks most time zones by offset from Coordinated Universal Time (UTC) (UTC-12:00 to UTC+12:00). [<SNTP server>]: SNTP server address or domain name.
Example	AT+CIPSNTPCFG=1,8,cn.pool.ntp.org OK
Notes	None

AT+CIPSNTPTIME? Command

Command	AT+CIPSNTPTIME?
Response	Returns the system SNTP time.
Function	Query SNTP time.
Parameter Description	None
Example	AT+CIPSNTPTIME? +CIPSNTPTIME:2025-7-6 20-5-4 OK
Notes	None

AT+CIPDOMAIN Command

Command	AT+CIPDOMAIN=<domain>[,<IP network>]
Response	Returns OK on success; returns error code on failure.
Function	Resolve domain name.
Parameter Description	<domain name>: Domain name to resolve. <IP address>: Resolved IP address.
Example	AT+CIPDOMAIN=www.baidu.com +CIPDOMAIN:183.240.99.169 OK
Notes	None

AT+CIPDNS Command

Command	AT+CIPDNS=<enable>[,<dns1>,<dns2>,<dns3>]
Response	Returns OK on success; returns error on failure.
Function	Set DNS server.

Parameter Description	<enable>: Set DNS: 0: Automatically obtain DNS Server from DHCP; 1: Specify DNS Server. <dns1>: When enable==1, specify DNS Server1. <dns2>: When enable==1, specify DNS Server2. <dns3>: When enable==1, specify DNS Server3.
Example	AT+CIPDNS=1,192.168.2.1,192.168.12.1 OK
Notes	None

8.6 BLE Command Set

Command	Function Description
AT+BLEINIT?	Query BLE role.
AT+BLEINIT	Set BLE role.
AT+BLENAME?	Query BLE device name.
AT+BLENAME	Set BLE device name.
AT+BLESCANPARAM?	Query BLE scan parameters.
AT+BLESCANPARAM	Set BLE scan parameters.
AT+BLESCAN	Start/Stop BLE scanning.
AT+BLEADDR?	Query BLE device address.
AT+BLESCANRSPDATA	Set BLE scan response data.
AT+BLEADVPARAM?	Query BLE advertising parameters.
AT+BLEADVPARAM	Set BLE advertising parameters.
AT+BLEADVDATA	Set BLE advertising data.
AT+BLEADVDATAEX?	Query automatic BLE advertising data.
AT+BLEADVDATAEX	Set automatic BLE advertising data.
AT+BLEADVSTART	Start BLE advertising.
AT+BLEADVSTOP	Stop BLE advertising.
AT+BLECONN	Establish a BLE connection.
AT+BLECONN?	Query device BLE connection.
AT+BLECONNPARAM?	Query BLE connection parameters.
AT+BLECONNPARAM	Set BLE connection parameters.
AT+BLEDISCONN	Disconnect an established BLE connection.
AT+BLEDATALEN	Set BLE connection data packet length.
AT+BLECFGMTU?	Query BLE connection MTU length.
AT+BLECFGMTU	Set BLE connection MTU length.
AT+BLEGATTSSRVCRE	Create GATTS service.
AT+BLEGATTSSRVSTART	Start a created GATTS service.
AT+BLEGATTSSRV?	View supported services on the device.
AT+BLEGATTSSCHAR?	View characteristics and descriptors of supported services.
AT+BLEGATTSENTFY	Server sends a notification message to Client.
AT+BLEGATTSEND	Server sends an Indication message to Client.
AT+BLEGATTCPRIMSRV	Client-side service discovery.
AT+BLEGATTCCCHAR	Client-side service characteristic discovery.
AT+BLEGATTCCR	Client sends data to Server.
AT+BLEGATTCD	Client reads data from Server.

AT+BLEINIT? Command

Command	AT+BLEINIT?
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Response	Returns the device BLE role.
Function	Query BLE role.
Parameter Description	None
Example	AT+BLEINIT? +BLEINIT:<role> OK
Notes	Returned role values: 0: Uninitialized; 1: Client role; 2: Server role.

AT+BLEINIT Command

Command	AT+BLEINIT=<role>
Response	Returns OK on success; returns error on failure.
Function	Set BLE role.
Parameter Description	role values: 0: Uninitialized; 1: Client role; 2: Server role.
Example	AT+BLEINIT=2 OK
Notes	Before using other BLE commands, call this command to initialize the device's BLE role. After role initialization, the role cannot be switched directly. To switch roles, you need to call the AT+RST command to restart the system first, then re-initialize the role. It is recommended to stop any ongoing advertising, scanning, and disconnect all connections before deregistering the BLE role (role==0).

AT+BLENAM? Command

Command	AT+BLENAM?
Response	Returns the BLE device name.
Function	Query BLE device name.
Parameter Description	None
Example	AT+BLENAM? +BLENAM:"BLE_Server01" OK
Notes	None

AT+BLENAM Command

Command	AT+BLENAME=<name>
Response	Returns OK on success; returns error on failure.
Function	Set BLE device name.
Parameter Description	name: BLE device name.
Example	AT+BLENAME=BLE_Server01 OK
Notes	The maximum length for the BLE device name is 18 characters.

AT+BLESCANPARAM? Command

Command	AT+BLESCANPARAM?
Response	Returns BLE scan parameters.
Function	Query BLE scan parameters.
Parameter Description	Returned BLE scan parameters: +BLESCANPARAM:<scan_type>,<own_addr_type>,<filter_policy>,<scan_interval>,<scan_window>
Example	AT+BLESCANPARAM? +BLESCANPARAM:0,0,0,100,50 OK
Notes	Device role must be set to Client first.

AT+BLESCANPARAM Command

Command	AT+BLESCANPARAM=<scan_type>,<own_addr_type>,<filter_policy>,<scan_interval>,<scan_window>
Response	Returns OK on success; returns error on failure.
Function	Set BLE scan parameters.
Parameter Description	<scan_type>: Scan type: 0: Passive scan; 1: Active scan. <own_addr_type>: Address type: 0: Public address; 1: Random address; 2: RPA public address; 3: RPA random address. <filter_policy>: Scan filter policy, setting not supported yet. <scan_interval>: Scan interval. Must be greater than or equal to <scan_window>. Range: [0x0004, 0x4000]. Actual scan interval = parameter * 0.625 ms, range: [2.5, 10240] ms. <scan_window>: Scan window. Must be less than or equal to <scan_interval>. Range: [0x0004, 0x4000]. Actual scan window = parameter * 0.625 ms, range: [2.5, 10240] ms.

Example	AT+BLEINIT=1 // First set role to Client AT+BLES SCANPARAM=0,0,0,100,50
Notes	Device role must be set to Client first.

AT+BLES SCAN Command

Command	AT+BLES SCAN=<enable>[,<interval>][,<filter_type>,<filter_param>]
Response	Returns OK on success; returns error on failure.
Function	Start/Stop BLE scanning.
Parameter Description	<enable>: 1: Start continuous scanning; 0: Stop continuous scanning. [<interval>]: Scan duration in seconds, setting not supported yet. [<filter_type>]: Filter option: 1: "MAC"; 2: "NAME". [<filter_param>]: Filter parameter, representing the peer device's MAC address or name. <addr>: BLE device address. <rssi>: Signal strength. <adv_data>: Advertising data. <scan_rsp_data>: Scan response data. <addr_type>: Advertising device address type.
Example	AT+BLEINIT=1 // Role Client AT+BLES SCAN=1 // Start scanning AT+BLES SCAN=0 // Stop scanning AT+BLES SCAN=1,3,1,"24:0A:C4:96:E6:88" // Start scan, filter by MAC address AT+BLES SCAN=1,3,2,"BK-AT" // Start scan, filter by device name
Notes	Device role must be set to Client first.

AT+BLEADDR? Command

Command	AT+BLEADDR?
Response	Returns the device's BLE address.
Function	Query BLE device address.
Parameter Description	None
Example	AT+BLEADDR? +BLEADDR:"15:09:42:8c:47:c8" OK
Notes	The device BLE address follows the Wi-Fi MAC address and is written during mass production.

AT+BLESCANRSPDATA Command

Command	AT+BLESCANRSPDATA=<scan_rsp_data>
Response	Returns OK on success; returns error on failure.
Function	Set BLE scan response data.
Parameter Description	<scan_rsp_data>: Scan response data, HEX string. For example, to set scan response data to 0x11 0x22 0x33 0x44 0x55, the command would be AT+BLESCANRSPDATA="1122334455".
Example	AT+BLEINIT=2 // Role Server AT+BLESCANRSPDATA="1122334455" OK
Notes	Device role must be set to Server first.

AT+BLEADVPARAM? Command

Command	AT+BLEADVPARAM?
Response	Returns BLE advertising parameters.
Function	Query BLE advertising parameters.
Parameter Description	Returned parameters: +BLEADVPARAM:<adv_int_min>,<adv_int_max>,<adv_type>,<channel_map>
Example	AT+BLEADVPARAM? +BLEADVPARAM:160,160,0,7 OK
Notes	Device role must be set to Server first.

AT+BLEADVPARAM Command

Command	AT+BLEADVPARAM=<adv_int_min>,<adv_int_max>,<adv_type>,<channel_map>
Response	Returns OK on success; returns error on failure.
Function	Set BLE advertising parameters.
Parameter Description	<adv_int_min>: Minimum advertising interval. Range: [0x0020, 0x4000]. Actual min interval = parameter * 0.625 ms, range: [20, 10240] ms. Must be ≤ <adv_int_max>. <adv_int_max>: Maximum advertising interval. Range: [0x0020, 0x4000]. Actual max interval = parameter * 0.625 ms, range: [20, 10240] ms. Must be ≥ <adv_int_min>.

	<adv_type>: 0: ADV_TYPE_IND, 1: ADV_TYPE_DIRECT_IND_HIGH, 2: ADV_TYPE_SCAN_IND, 3: ADV_TYPE_NONCONN_IND, 4: ADV_TYPE_DIRECT_IND_LOW. <channel_map>: Advertising channels: 1: ADV_CHNL_37, 2: ADV_CHNL_38, 4: ADV_CHNL_39, 7: ADV_CHNL_ALL.
Example	AT+BLEINIT=2 // Role Server AT+BLEADVPARAM=160,160,0,7 OK
Notes	Device role must be set to Server first.

AT+BLEADVDATA Command

Command	AT+BLEADVDATA=<adv_data>
Response	If successful, returns OK; if failed, returns error.
Function	Sets BLE broadcast data.
Parameter description	<adv_data>: Broadcast data is in HEX string format. For example, if you want to set the "Local Name" of the broadcast data to "7200_BLE", the command is AT+BLEADVDATA="0909373230305F424C45".
Example	AT+BLEINIT=2 //Role Server AT+BLEADVDATA="0909373230305F424C45" OK
Notes	If the broadcast data has been previously set using the command AT+BLEADVDATAEX=<dev_name>,<uuid>,<manufacturer_data>,<include_power> then the broadcast data set by this command will overwrite it. The device must first be set as the Server role.

AT+BLEADVDATAEX? Command

Command	AT+BLEADVDATAEX?
Response	Returns automatic BLE advertising data.
Function	Query BLE automatic advertising data.
Parameter Description	Returned parameters: +BLEADVDATAEX:<dev_name>,<uuid>,<manufacturer_data>,<include_power>
Example	AT+BLEADVDATAEX? +BLEADVDATAEX:"BLE_Server01","a002","1122334455",1 OK
Notes	Device role must be set to Server first.

AT+BLEADVDATAEX Command

Command	AT+BLEADVDATAEX=<dev_name>,<uuid>,<manufacturer_data>,<include_power>
Response	Returns OK on success; returns error on failure.
Function	Set BLE automatic advertising data.
Parameter Description	<dev_name>: String parameter, representing the device name. If the device name is set to "just-test", the command is AT+BLEADVSTARTEX="just-test",<uuid>,<manufacturer_data>,<include_power>; <uuid>: String parameter. If you want to set the UUID to "0xA002", the command is AT+BLEADVSTARTEX=<dev_name>,"A002",<manufacturer_data>,<include_power>; <manufacturer_data>: Manufacturer data, a HEX string. If you want to set the manufacturer data to "0x11 0x22 0x33 0x44 0x55", the command is AT+BLEADVSTARTEX=<dev_name>,<uuid>,"1122334455",<include_power>; <include_power>: Not supported for setting.
Example	AT+BLEINIT=2 // Role Server AT+BLEADVDATAEX="BLE_Server01","a002","1122334455",1
Notes	Device role must be set to Server first. If advertising data was previously set using the AT+BLEADVDATA command, it will be overwritten by the advertising data set by this command.

AT+BLEADVSTART Command

Command	AT+BLEADVSTART
Response	Returns OK on success; returns error on failure.
Function	Start BLE advertising.
Parameter Description	None
Example	AT+BLEINIT=2 // Role Server AT+BLENAME=BLE_Server01 // Set device name AT+BLESCANRSPDATA="1122334455" // Set scan response data AT+BLEADVPARAM=160,160,0,7 // Set advertising parameters AT+BLEADVDATAEX="BLE_Server01","a002","1122334455",1 //Set auto advertising data AT+BLEADVSTART // Start advertising AT+BLEADVSTOP // Stop advertising
Notes	Device role must be set to Server first. If advertising parameters are not set using AT+BLEADVPARAM, default parameters are used. If advertising data is not set, a zero-packet is sent.

	If advertising data is set using AT+BLEADVDATA, that data packet is sent. If advertising data is set using AT+BLEADVDATAEX, that data packet is sent. After starting BLE advertising, it will continue until a BLE connection is established, after which advertising stops.
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AT+BLEADVSTOP Command

Command	AT+BLEADVSTOP
Response	Returns OK on success; returns error on failure.
Function	Stop BLE advertising.
Parameter Description	None
Example	AT+BLEINIT=2 // Role Server AT+BLENAME=BLE_Server01 // Set device name AT+BLESCANRSPDATA="1122334455" // Set scan response data AT+BLEADVPARAM=160,160,0,7 // Set advertising parameters AT+BLEADVDATAEX="BLE_Server01","a002","1122334455",1 //Set auto advertising data AT+BLEADVSTART // Start advertising AT+BLEADVSTOP // Stop advertising
Notes	Device role must be set to Server first.

AT+BLECONN Command

Command	AT+BLECONN=<conn_index>,<remote_address>[,<addr_type>,<timeout>]
Response	Returns OK on success; returns error on failure.
Function	Establish a BLE connection.
Parameter Description	<conn_index>: BLE connection number, range: [0, 2]. <remote_address>: Peer BLE device address. [<addr_type>]: Advertising device address type, default 0: 0: Public Address, 1: Random Address. [<timeout>]: Setting not supported yet, default 5s.
Example	AT+BLEINIT=1 // Role Client AT+BLENAME=BLE_Client01 // Set device name AT+BLESCANPARAM=0,0,0,100,50 // Set scan parameters AT+BLESCAN=1 // Start scanning AT+BLECONN=0,1509428c47c8 // Connect to Server device with address "15:09:42:8c:47:c8"

	AT+BLESCAN=0 // After successful connection, it's recommended to stop scanning
注意事项	First, set the device role to "Client" and start the scan. For the connected Server device, you need to configure the role service, broadcast parameters, and enable broadcasting. Refer to the following settings: AT+BLEINIT=2 // Role: Server AT+BLENAME=BLE_Server01 // Set device name AT+BLEGATTSSRVCRE // Create service AT+BLEGATTSSRVSTART // Start all services AT+BLESCANRSPDATA="1122334455" // Set scan response data AT+BLEADVPARAM=160,160,0,7 // Set broadcast parameters AT+BLEADVDATAEX="BLE_Server01","a002","1122334455",1 // Set automatic broadcast data AT+BLEADVSTART // Start broadcasting

AT+BLECONN? Command

Command	AT+BLECONN?
Response	Returns the device's BLE connection information.
Function	Query the device's BLE connection.
Parameter Description	Returned connection info format: +BLECONN:<conn_index>,<remote_address>
Example	AT+BLECONN? +BLECONN:0,"15:09:42:8c:47:c8" OK
Notes	Device role must be set to Client and a BLE connection must be successfully established.

AT+BLECONNPARAM? Command

Command	AT+BLECONNPARAM?
Response	Returns BLE connection parameters.
Function	Query BLE connection parameters.
Parameter Description	Returned connection parameters format: +BLECONNPARAM:<conn_index>,<min_interval>,<max_interval>,<cur_interval>,<latency>,<timeout>
Example	AT+BLECONNPARAM? +BLECONNPARAM:0,87,87,87,0,500 OK
Notes	Device role must be set to Client and a BLE connection must be successfully established.

AT+BLECONNPARAM Command

Command	AT+BLECONNPARAM=<conn_index>,<min_interval>,<max_interval>,<latency>,<timeout>
Response	Returns OK on success; returns error on failure.
Function	Set BLE connection parameters.
Parameter Description	<conn_index>: BLE connection number, range: [0, 2]. <min_interval>: Minimum connection interval. Must be $\leq$ <max_interval>. Range: [0x0006, 0x0C80]. Actual min interval = parameter * 1.25 ms, range: [7.5, 4000] ms. <max_interval>: Maximum connection interval. Must be $\geq$ <min_interval>. Range: [0x0006, 0x0C80]. Actual max interval = parameter * 1.25 ms, range: [7.5, 4000] ms. <latency>: Latency, range: [0x0000, 0x01F3]. <timeout>: Timeout, range: [0x000A, 0x0C80]. Actual timeout = parameter * 10 ms, range: [100, 32000] ms.
Example	AT+BLECONNPARAM=0,160,160,0,500 OK
Notes	Device role must be set to Client and a BLE connection must be successfully established.

AT+BLEDISCONN Command

Command	AT+BLEDISCONN=<conn_index>
Response	Returns OK on success; returns error on failure.
Function	Disconnect an established BLE connection.
Parameter Description	<conn_index>: BLE connection number, range: [0, 2].
Example	AT+BLEDISCONN=0 OK +BLEDISCONN:0,"15:09:42:8c:47:c8"
Notes	Device role must be set to Client and a BLE connection must be successfully established.

AT+BLEDATALEN Command

Command	AT+BLEDATALEN=<conn_index>,<pkt_data_len>
Response	Returns OK on success; returns error on failure.
Function	Set BLE connection data packet length.

Parameter Description	<conn_index>: BLE connection number, range: [0, 2]. <pkt_data_len>: Packet data length, range: [0x001B, 0x00FB].
Example	AT+BLEDATALEN=0,64 OK
Notes	Device role must be set to Client and a BLE connection must be successfully established.

AT+BLECFGMTU? Command

Command	AT+BLECFGMTU?
Response	Returns the BLE connection MTU length.
Function	Query BLE connection MTU length.
Parameter Description	Returned message format: +BLECFGMTU:<conn_index>,<mtu_size>
Example	AT+BLECFGMTU? +BLECFGMTU:0,23 OK
Notes	Device role must be set to Client and a BLE connection must be successfully established.

AT+BLECFGMTU Command

Command	AT+BLECFGMTU=<conn_index>,<mtu_size>
Response	Returns OK on success; returns error on failure.
Function	Set BLE connection MTU length.
Parameter Description	<conn_index>: Bluetooth LE connection number, range: [0, 2]. <mtu_size>: Setting not supported yet.
Example	AT+BLECFGMTU=0,64 OK
Notes	Device role must be set to Client and a BLE connection must be successfully established. The actual MTU length needs to be negotiated. A successful response only indicates an attempt to negotiate the MTU length, so the set length may not take effect. It is recommended to query the actual MTU length using AT+BLECFGMTU?.

AT+BLEGATTSSRVCRE Command

Command	AT+BLEGATTSSRVCRE
Response	Returns OK on success; returns error on failure.

Function	Create GATTS service.
Parameter Description	None
Example	AT+BLEGATTSSRVCRE OK
Notes	Before creating a service for BLE Server, the Service.bin file must be burned into the BLE SERVER partition. This is done during device mass production. With device role set to Server, this command can be called to create the service. Services must be created before establishing a BLE connection.

AT+BLEGATTSSRVSTART Command

Command	AT+BLEGATTSSRVSTART[=<srv_index>]
Response	Returns OK on success; returns error on failure.
Function	Start a created GATTS service.
Parameter Description	srv_index: Specified service number.
Example	AT+BLEGATTSSRVSTART // Start all created services AT+BLEGATTSSRVSTART=2 // Start service number 2
Notes	Device role must be set to Server and services must be created first. Services must be created and started before establishing a BLE connection.

AT+BLEGATTSSRV? Command

Command	AT+BLEGATTSSRV?
Response	Returns the services supported by the device.
Function	View supported services on the device.
Parameter Description	Returned parameters: +BLEGATTSSRV:<srv_index>,<start>,<srv_uuid>,<srv_type>
Example	AT+BLEGATTSSRV? //The following are the services supported by the device +BLEGATTSSRV:1,1,0xed2b4e3a2820492f9507df165285e831,1 +BLEGATTSSRV:2,1,0xaabb,1 OK
Notes	Device role must be set to Server and services must be created first.

AT+BLEGATTCHAR? Command

Command	AT+BLEGATTCHAR?
Response	Returns the characteristics and descriptors of supported services.
Function	View characteristics and descriptors of supported services.
Parameter Description	// Description of the returned feature format: +BLEGATTCHAR:"char",<srv_index>,<char_index>,<char_uuid>,<char_prop> +BLEGATTCHAR:"desc",<srv_index>,<char_index>,<desc_index> <srv_index>: Service number, starting from 1. <char_index>: Characteristic number within service, starting from 1. <char_uuid>: UUID of the characteristic. <char_prop>: Properties of the characteristic. <desc_index>: Characteristic descriptor number. <desc_uuid>: UUID of the characteristic descriptor.
Example	AT+BLEGATTCHAR? //The following are the services supported by the device +BLEGATTCHAR:"char",1,1,0x31e8855216df07952f4920283b4e2bed,0x8008 // The 1st service and 2nd feature below can be used to send notification messages to the client. +BLEGATTCHAR:"char",1,2,0x31e8855216df07952f4920283c4e2bed,0x8010 +BLEGATTCHAR:"desc",1,2,1 +BLEGATTCHAR:"char",1,3,0x01ff,0x2 // The second service of number 2 and the first feature of number 1 can be used to send the Indicate message to the client. +BLEGATTCHAR:"char",2,1,0x02ff,0x20 +BLEGATTCHAR:"desc",2,1,1 +BLEGATTCHAR:"char",2,2,0x03ff,0x2 OK
Notes	Device role must be set to Server and services must be created first

AT+BLEGATTSENTFY command

Command	AT+BLEGATTSENTFY=<conn_index>,<srv_index>,<char_index>,<length>
Response	When the command is executed successfully, it will prompt ">>" to enter a string of length (no more than 64 characters). After entering the specified length data, immediately follow it with "/r/n" to indicate the end.

Function	Sever sends a notification message to Client
Parameter Description	<conn_index>: BLE connection identifier, range: [0, 2]. <srv_index>: Service identifier supported by the device, see AT+BLEGATTCHAR? Explanation. <char_index>: The characteristic number that can be used to send notification messages to the Client. Refer to AT+BLEGATTCHAR? Explanation. <length>: The length of the transmitted string. It indicates that the length of the input string should be consistent.
Example	<pre>// Instructions to be executed in advance on the Server side before sending a notification message to the Client: AT+BLEINIT=2 // Role: Server AT+BLENAME=BLE_Server01 // Set the device name AT+BLEGATTSSRVCRE // Create a service AT+BLEGATTSSRVSTART // Start all services AT+BLESCANRSPDATA="1122334455" // Set the scan response data AT+BLEADVPARAM=160,160,0,7 // Set the advertising parameters AT+BLEADVDATAEX="BLE_Server01","a002","1122334455",1 // Set the automatic advertising data AT+BLEADVSTART // Start advertising. // After the Client successfully connects, configures the connection parameters, and discovers the service characteristics, send a notification message to the Client. AT+BLEGATTSENTFY=0,1,2,50 // Send a notification message using Service No. 1 and Characteristic No. 2 >123456789012345678901234567890123456789012345678901234567890 OK</pre>
Notes	<pre>// Instructions to be pre - executed on the Client side before receiving notification messages from the Server: AT+BLEINIT=1 // Role: Server AT+BLENAME=BLE_Client01 // Set the device name AT+BLESCANPARAM=0,0,0,100,50 // Set the scanning parameters AT+BLESCAN=1 // Start scanning AT+BLECONN=0,1509428c47c8 // Connect to the Server device with the address "15:09:42:8c:47:c8" AT+BLESCAN=0 // It is recommended to turn off scanning after a successful connection AT+BLECONNPARAM=0,160,160,0,500 // Configure the connection parameters AT+BLEDATALEN=0,64 // Set the data packet length to 64 AT+BLECFGMTU=0,64 // Set the BLE connection MTU length AT+BLEGATTCPRIMSRV=0 // Set the service discovery AT+BLEGATTCCCHAR=0,1 // Discover the characteristics of Service No. 1 AT+BLEGATTCCCHAR=0,2 // Discover the characteristics of Service No. 2</pre>

	AT+BLEGATTCCHAR=0,3 // Discover the characteristics of Service No. 3 AT+BLEGATTCCHAR=0,4 // Discover the characteristics of Service No. 4 // Notification message received from the Server: +BLEGATTSNTFY:0,50:12345678901234567890123456789012345678901234567890
--	--

AT+BLEGATTSIND Command

Command	AT+BLEGATTSIND=<conn_index>,<srv_index>,<char_index>,<length>
Response	On successful command execution, prompts with ">" to input a string of <length> characters (max 64). Input the specified length of data followed by "rn" to indicate the end.
Function	Server sends an Indication message to Client.
Parameter Description	<conn_index>: BLE connection number, range: [0,2]. <srv_index>: Supported service number, see AT+BLEGATTSCCHAR. <char_index>: Characteristic number usable for sending notifications to Client , see AT+BLEGATTSCCHAR?。 <length>: Length of the data string to transmit. The length of the prompted input string must match。
Example	Before sending the Indicate message to the Client, the Server needs to execute the following instructions in advance as per the description in AT+BLEGATTSNTFY After Client connects, configures parameters, and discovers service characteristics, send Indication to Client。 AT+BLEGATTSIND=0,2,1,50 / Send an Indicate message using service 2 and feature 1 >12345678901234567890123456789012345678901234567890 OK
Notes	Pre-execution commands on Client side before receiving Indications from Server, refer to AT+BLEGATTSNTFY notes..... Received Indication from Server: +BLEGATTSIND:0,50:123456789012345678901234567890123456789012345678900

AT+BLEGATTCPRIMSRV Command

Command	AT+BLEGATTCPRIMSRV=<conn_index>
Response	On success, returns discovered services.

Function	Client-side service discovery.
Parameter Description	Returned service message format: +BLEGATTCPRIMSRV:<conn_index>,<srv_index>,<srv_uuid>,<srv_type> <conn_index>: BLE connection number, range: [0, 2]. <srv_index>: Service number. <srv_uuid>: UUID of the service. <srv_type>: Service type: 0: Secondary service; 1: Primary service.
Example	<pre>AT+BLEGATTCPRIMSRV=0// Services discovered by the device (fixed content) +BLEGATTCPRIMSRV:0,1,0x1800,1 +BLEGATTCPRIMSRV:0,2,0x1801,1 +BLEGATTCPRIMSRV:0,3,0xed2b4e3a2820492f9507df165285e831,1 +BLEGATTCPRIMSRV:0,4,0xaabb,1 OK</pre>
Notes	Device role must be set to Client and a BLE connection must be successfully established.

AT+BLEGATTCCCHAR Command

Command	AT+BLEGATTCCCHAR=<conn_index>,<srv_index>
Response	On success, returns characteristics of the specified service
Function	Client-side service characteristic discovery
Parameter Description	Message format for returning specified service characteristics: // Message format for discovered characteristics: +BLEGATTCCCHAR:"char",<conn_index>,<srv_index>,<char_index>,<char_uuid>,<char_prop> // The message format of the discovered descriptors: +BLEGATTCCCHAR:"desc",<conn_index>,<srv_index>,<char_index>,<desc_index>,<desc_uuid> <conn_index>: BLE connection identifier, range: [0, 2]. <srv_index>: Service index. <char_index>: Index of service characteristic. <char_uuid>: UUID of service characteristic. <char_prop>: Attribute of service characteristic. <desc_index>: Descriptor index of characteristic. <desc_uuid>: UUID of descriptor.
Example	<pre>// Discover the characteristics of Service 1: AT+BLEGATTCCCHAR=0,1 +BLEGATTCCCHAR:"char",0,1,1,0x002a00,0x2 //readable +BLEGATTCCCHAR:"char",0,1,2,0x002a01,0x2 //readable</pre>

	OK // Discover the characteristics of Service No. 2: AT+BLEGATTCCHAR=0,2 +BLEGATTCCHAR:"char",0,2,1,0x002a05,0x20 +BLEGATTCCHAR:"desc",0,2,1,1,0x002902,0x2 +BLEGATTCCHAR:"char",0,2,2,0x002b29,0xa +BLEGATTCCHAR:"char",0,2,3,0x002b2a,0x2 //readable +BLEGATTCCHAR:"char",0,2,4,0x002b3a,0x2 //readable OK // Discover the characteristics of Service 3: AT+BLEGATTCCHAR=0,3 //Service 3 can write Feature 1, which is used for the Client to send messages to the Server. +BLEGATTCCHAR:"char",0,3,1,0x0000000000000000ed2b4e3b2820492f,0x8 //writeable +BLEGATTCCHAR:"char",0,3,2,0x0000000000000000ed2b4e3c2820492f,0x10 +BLEGATTCCHAR:"desc",0,3,2,1,0x002902,0x0 +BLEGATTCCHAR:"char",0,3,3,0x00ff01,0x2 //readable OK // Discover the characteristics of Service No. 4: AT+BLEGATTCCHAR=0,4 +BLEGATTCCHAR:"char",0,4,1,0x00ff02,0x20 +BLEGATTCCHAR:"desc",0,4,1,1,0x002902,0x10 +BLEGATTCCHAR:"char",0,4,2,0x00ff03,0x2 //readable OK
Notes	First, the device must be set as the Client and a successful BLE connection must be established.

AT+BLEGATTCWR Command

Command	AT+BLEGATTCWR=<conn_index>,<srv_index>,<char_index><length>
Response	On successful command execution, prompts with ">" to input a string of <length> characters (max 64). Input the specified length of data followed by "rn" to indicate the end.
Function	Client sends data to Server.
Parameter Description	<conn_index>: BLE connection number, range: [0, 2]. <srv_index>: Service number discovered for the connection, see AT+BLEGATTCPRIMSRV. <char_index>: Characteristic number discovered for the service that is writable, see AT+BLEGATTCCHAR. <length>: Length of the data string to transmit. The length of the prompted input string must match.

Example	<pre>// Pre-execution commands on Client side before sending data to Server, refer to AT+BLEGATTSENTFY notes. // After Client connects, configures parameters, and discovers service characteristics, send data to Server. AT+BLEGATTCWR=0,3,1,50 // Use service 3, characteristic 1 to send data >12345678901234567890123456789012345678901234567890 OK</pre>
Notes	<pre>// Pre-execution commands on Server side before receiving data from Client, refer to AT+BLEGATTSENTFY notes. ... // Received data from Client: +ClientWRITE:0,50:123456789012345678901234567890123456789012 34567890</pre>

AT+BLEGATTCRD Command

Command	AT+BLEGATTCRD=<conn_index>,<srv_index>,<char_index>
Response	On successful command execution, returns the characteristic value read from the Server.
Function	Client reads data from Server.
Parameter Description	<conn_index>: BLE connection number, range: [0, 2]. <srv_index>: Service number discovered for the connection, see AT+BLEGATTCPRIMSRV. <char_index>: Characteristic number discovered for the service that is readable, see AT+BLEGATTCCHAR.
Example	<pre>// Pre-execution commands on Client side before sending read request to Server, refer to AT+BLEGATTSENTFY notes. // After Client connects, configures parameters, and discovers service characteristics, send read request to Server. AT+BLEGATTCRD=0,3,3 // Read characteristic 3 of service 3 +BLEGATTCRD:0,03:12 34 56 // Returns 3 bytes read: 0x12 0x34 0x56.</pre>
Notes	<pre>// Pre-execution commands on Server side before receiving read request from Client, refer to AT+BLEGATTSENTFY notes. ... // Received read request from Client: +READ:0,0,7,0,00</pre>

8.7 OTA Command Set

Command	Function Description
AT+OTA	Firmware OTA update.

AT+OTA Command

Command	AT+OTA=<ota_image_http_url>
Response	On successful execution, downloads firmware from HTTP Server, burns firmware, and restarts the system.
Function	Firmware OTA update.
Parameter Description	ota_image_http_url: URL of the OTA firmware.
Example	<pre> AT+OTA=http://192.168.100.19:65534/HM-WF7201_OTA_v2.0.rbl OK RT-Thread OTA package(V0.2.8-beken-1133282d-20220604) initialize success. OTA firmware(app) upgrade(2M.1220->2M.1220) startup. OTA Write: [==>] 3% OTA Write: [=====] 100% ota_erase_dl_rbl erase:132000. // System restart..... </pre>
Notes	1、 Need to set up an HTTP SERVER and load the OTA firmware. 2、 The device must be configured in STA or STA+AP mode, connect to a wireless router, and be able to ping the server or PC running the HTTP SERVER.

OTA Firmware Upgrade Operation Example:

- 1、 PC connects to Router A, IP address is 192.168.100.19.
- 2、 Set up an HTTP SERVER on the PC and load the OTA firmware. This example uses the "HFS" tool. It is recommended to turn off the PC firewall to avoid blocking access to the HTTP SERVER. See Figure 8-1.

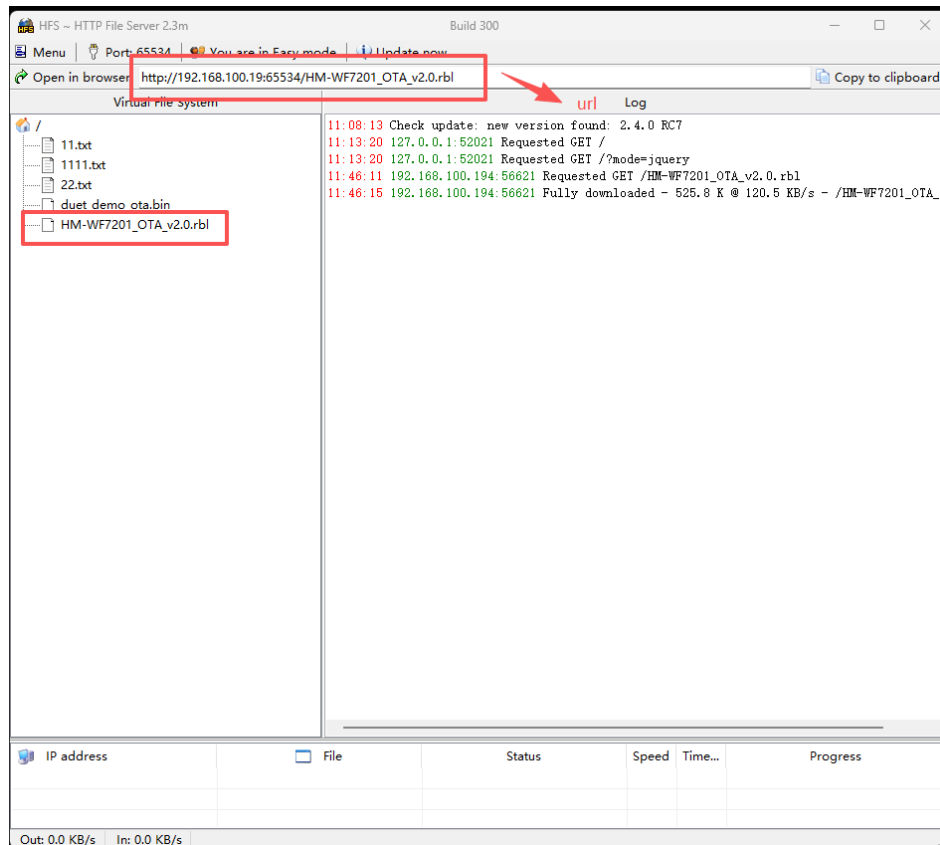


Figure 8-1 Using the "HFS" tool to build an HTTP SERVER

3、The device also connects to Router A. Execute the following AT commands:

- o AT+RESTORE // Restore to factory settings to eliminate any other adverse factors
- o AT+CWDHCP=0,2 // Disable DHCP SERVER
- o AT+CWDHCP=1,1 // Enable DHCP CLIENT
- o AT+CWMODE=1 // Set device Wi-Fi to STA mode
- o AT+CWJAP=ap_ssid,ap_password // Connect to Router A
- o AT+CIPSTA? // Check device IP address/mask/gateway, confirm successful connection to Router A
- o AT+PING=192.168.100.19 // Ensure the device can ping the PC

4、Trigger OTA upgrade. After the device firmware upgrades and restarts, check the firmware version number to ensure the upgrade was successful

AT+OTA = http://192.168.100.19:65534/HM-WF7201_OTA_v2.0.rbl // Refer to Figure 8-1
AT+GMR? // Check the firmware version number, as shown in Figure 8-2



Figure 8-2 View Firmware Version Number

9 Hardware Design Considerations

1. It is recommended to use a DC regulated power supply for the module, with power ripple as low as possible. The module must be reliably grounded. Ensure correct connection of power supply polarity; reverse connection may cause permanent damage to the module.
2. Check the power supply to ensure it is within the recommended voltage range. Exceeding the maximum value can cause permanent damage. Ensure power supply stability; voltage should not fluctuate significantly or frequently.
3. When designing the power supply circuit for the module, it is recommended to reserve a margin of over 30% to ensure long-term stable operation of the whole device. The module should be placed as far away as possible from components with significant electromagnetic interference, such as power supplies, transformers, and high-frequency traces.
4. High-frequency digital traces, high-frequency analog traces, and power traces must avoid passing under the module. If it is unavoidable to route traces under the module, assuming the module is soldered on the Top Layer, the Top Layer area under the module contact points should be grounded (full copper pour with good grounding), and traces should be kept close to the digital part of the module and routed on the Bottom Layer.
5. Assuming the module is soldered or placed on the Top Layer, arbitrarily routing traces on the Bottom Layer or other layers is also incorrect and can affect the module's spurious emissions and reception sensitivity to varying degrees.
6. If there are components with significant electromagnetic interference near the module, it can greatly affect the module's performance. It is recommended to keep a proper distance from the module based on the interference intensity, and if conditions permit, appropriate isolation and shielding should be implemented.
7. If there are traces with significant electromagnetic interference (high-frequency digital, high-frequency analog, power traces) near the module, it can greatly affect the

module's performance. It is recommended to keep a proper distance from the module based on the interference intensity, and if conditions permit, appropriate isolation and shielding should be implemented.

8. If communication lines use 5V logic levels, a level shifter circuit must be used.
9. Keep as far away as possible from some TTL protocols operating in the 2.4 GHz band, such as USB 3.0.
10. Refer to Figure 9.1 and Figure 9.2 for module antenna layout and routing suggestions。

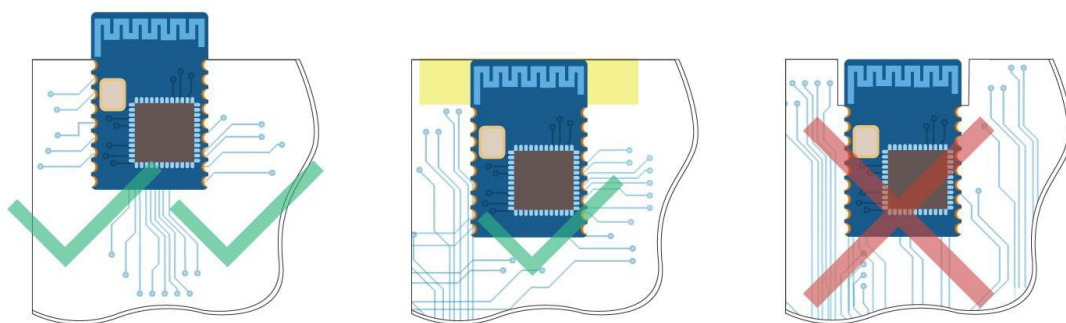


Figure9.1 PCB Layout Suggestion Diagram

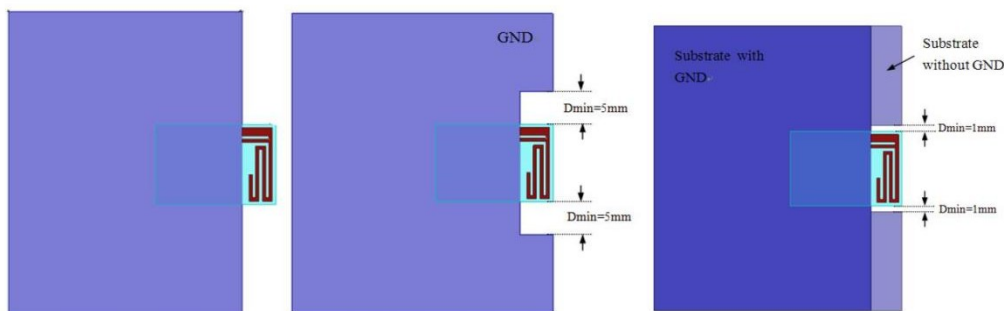


Figure9.2 PCB Placement Suggestion Diagram

10 Frequently Asked Questions

10.1 Unsatisfactory Transmission Distance

1. Communication distance will decrease when there is a line-of-sight obstruction. Temperature, humidity, and co-channel interference can increase the data packet loss rate. The ground absorbs and reflects radio waves, so testing near the ground yields poor results.
2. Seawater strongly absorbs radio waves, so testing near the sea yields poor results.
3. Metal objects near the antenna, or placing the module inside a metal enclosure, will severely attenuate the signal.
4. Incorrect power register settings or an excessively high over-the-air rate (higher rate leads to shorter distance).
5. Power supply voltage below the recommended value at room temperature; lower voltage results in lower transmission power.
6. Poor matching between the antenna and the module, or poor quality of the antenna itself.

10.2 Susceptibility to Damage and Abnormal Failure

1. Check the power supply to ensure it is within the recommended voltage range. Exceeding the maximum value can cause permanent damage. Ensure power supply stability; voltage should not fluctuate significantly or frequently.
2. Ensure anti-static measures are taken during installation and use, as high-frequency devices are sensitive to electrostatic discharge.
3. Ensure the humidity during installation and use is not too high, as some components are sensitive to moisture. It is not recommended to use the module at extremely high or low temperatures unless specifically required.

10.3 High Bit Error Rate

1. There may be co-channel signal interference nearby. Move away from the interference source or change the frequency/channel to avoid the interference.

2. An unstable power supply can also cause data errors. Ensure the power supply is reliable.
3. Poor quality or excessively long extension cables or feed lines can also lead to a high bit error rate.

11 Reflux Welding Conditions

1. Heating method: conventional convection or IR convection;
2. Number of allowable reflow welding: 2 times, based on the following reflow welding (conditions) (see the figure below);
3. Temperature curve: reflow welding should follow the following temperature curve (see the figure below);
4. Maximum temperature: 245°C。

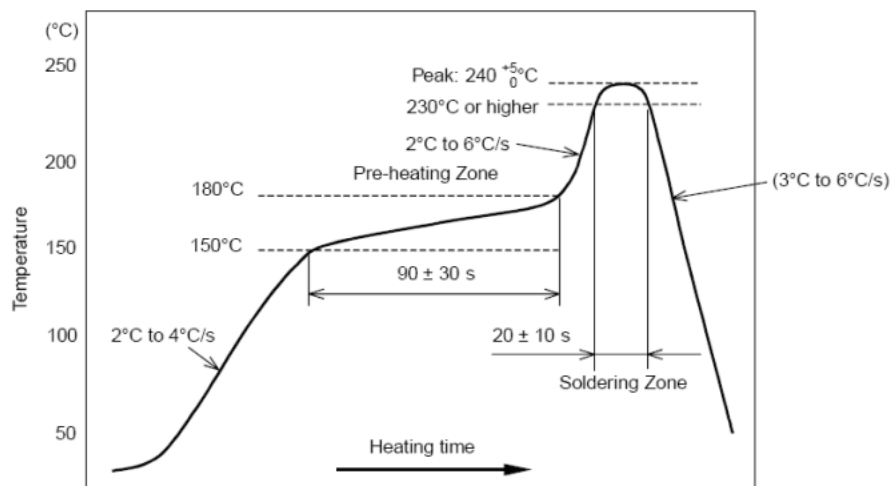


Figure 11.1 Welding heat resistance temperature curve of components (welding points)

12 Electrostatic Discharge Warning

The module is damaged due to static release and it is recommended that all modules be treated under the following 3 precautions:

1. Must follow the anti-static measures, can not hold the module.
2. The module must be placed in a placement area that can prevent static electricity.
3. The antistatic circuit at the high-voltage input or high-frequency input should be considered in the product design.

Electrostatics may result in subtle performance degradation to the entire device failure. Because very small parameter changes may cause the device to not meet the value limit of its certification requirements, the module will be more vulnerable to damage.

13 Document Change Record

Table 13.1 Document change record

Document version	Change description	Date updated
V1.0	First release	2025.12.15

14 Contact Information

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